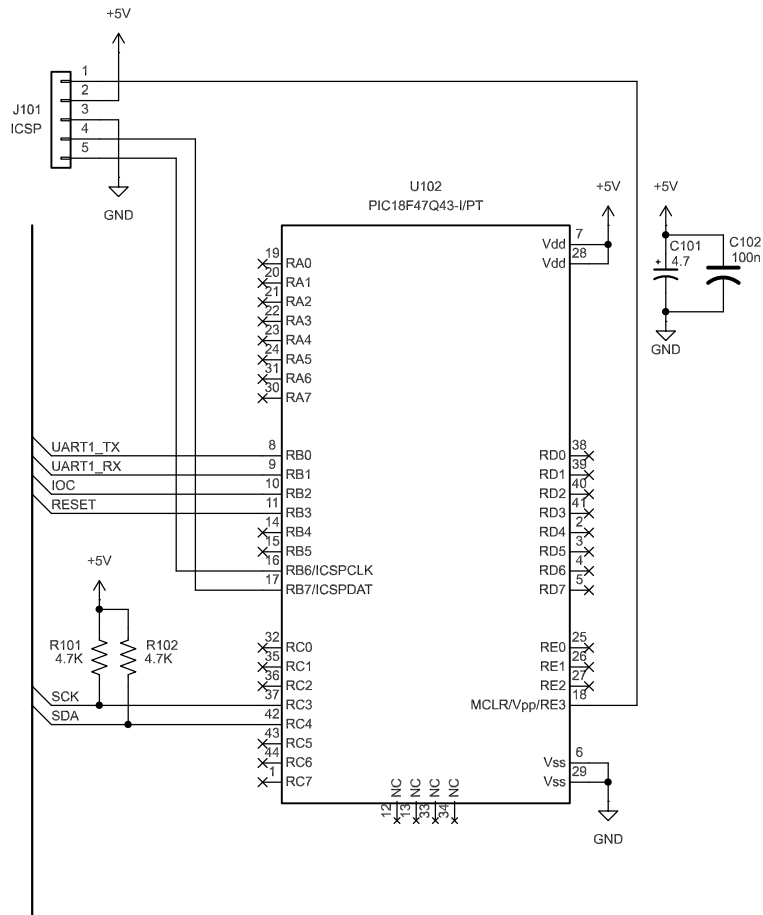
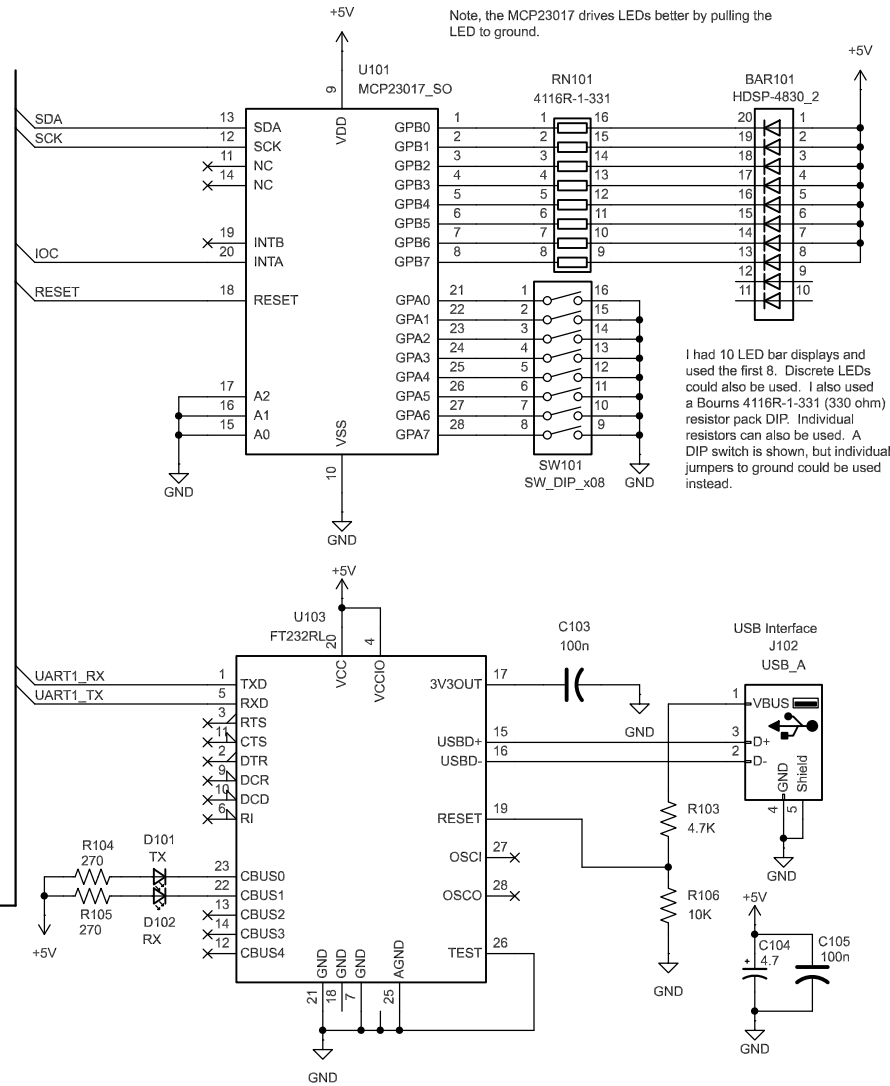




This example circuit uses an I/O expander (MCP23017) connected via I2C interface with interrupt on change enabled. It could also be polled periodically for changes.

Also notice that the I2C lines (SCK and SDA) have external 4.7K pull-up resistors installed. This is because we are using a bit-bang interface with the normal I/O pins and weak pull-ups will not be enough. Later, in another module, we will switch to using the internal I2C peripheral module which provides adequate pull-ups and these resistors will be deleted.



Note, the MCP23017 drives LEDs better by pulling the LED to ground.

I had 10 LED bar displays and used the first 8. Discrete LEDs could also be used. I also used a Bourns 4116R-1-331 (330 ohm) resistor pack DIP. Individual resistors can also be used. A DIP switch is shown, but individual jumpers to ground could be used instead.